

Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is transforming the healthcare industry by improving diagnosis, treatment planning, patient monitoring, and hospital management. Machine learning models help doctors detect diseases such as cancer, diabetes, and heart conditions from medical images and clinical data. Natural Language Processing (NLP) enables healthcare providers to analyze electronic health records, summarize medical reports, and extract important patient information. AI-powered virtual assistants support appointment scheduling and patient queries, while predictive analytics helps identify high-risk patients before complications occur. Although AI offers significant benefits, challenges such as data privacy, security, ethical concerns, and model bias must be addressed. With responsible implementation, AI has the potential to improve healthcare quality, reduce costs, and enhance patient outcomes.